

Varun Woodi Raghavendra

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(Open to Relocate)

SUMMARY

Results-driven **ML Engineer** with **3+ years** of experience in **AI/ML**. Proficient in **Python, SQL, and AWS**, with expertise in **TensorFlow, PyTorch, NLP, and Computer Vision**. Designed scalable **AI/ML** pipelines, optimizing **Deep Learning** models for automation and predictive analytics. Improved model accuracy and reduced manual effort by **40%** through automation. Built **predictive models** enhancing decision-making in **finance, healthcare, and e-commerce**. Developed real-time analytics dashboards, increasing operational efficiency. **Actively** Seeking opportunities to lead **ML** projects and optimize **AI** solutions at scale.

AWARDS, RECOGNITION, CERTIFICATIONS

- [Data Analyst Certification](#)
- [GenAI with LangChain and HuggingFace](#)
- [MLOps Bootcamp with End-to-end Projects](#)
- Data Science Certification from ExcelR
- State cricket player for Karnataka State, India

WORK EXPERIENCE

Health Innovators

Remote, US

ML Engineer

Sep 2024 - Present

An LLM-powered healthcare chatbot using LLM, providing real-time, context-aware responses. Ensures HIPAA compliance while delivering personalized medical support to patients.

- Developed an AI-powered healthcare chatbot using the **MedAlpaca** model, leveraging **NLP, Named Entity Recognition (NER), and predictive analytics** to automate real-time patient interactions while ensuring HIPAA compliance.
- Designed and optimized real-time data retrieval and storage systems with **AWS DynamoDB**, ensuring efficient handling of structured patient records, medical FAQs, and chatbot logs for context-aware AI interactions and improved patient support.
- Leveraged **R** for advanced data analysis, generating insights to optimize chatbot performance and enhance patient interaction strategies.
- Utilized **Python** to develop the Caddy chatbot, enabling real-time patient support, data processing, and scalable backend solutions.
- Implemented **CI/CD pipelines** to automate chatbot deployment, ensuring faster updates, continuous integration, and streamlined development workflows.
- Applied **machine learning** techniques to enhance chatbot response accuracy, improving intent recognition and user engagement through continuous model training.
- Developed **AI-based** solutions for the Caddy chatbot, enhancing patient interaction with intelligent responses and real-time processing.
- Leveraged **AWS SageMaker** for scalable model training and inference, using **MLflow** for tracking, versioning, and performance analysis.
- Conducted **A/B testing** to compare fine-tuned LLM models on accuracy, latency, and failure rates, using historical medical query datasets to optimize chatbot performance before deployment.
- Developed a scalable Web UI using **Streamlit**, hosted on **AWS EC2**, enabling real-time chatbot interactions while integrating with a database for data storage and retrieval.
- Monitored and analyzed model performance using MLflow's tracking and UI, selecting the best-performing version of the **LLM model** through model comparison experiments, improving response times to under 2 seconds, and achieving **90%** accuracy in medical Q&A

INFOSYS

Analyst/Systems Engineer

Client: Medtronic

Bangalore, India

May 2021 - Aug 2022

Developed a predictive analytics solution to forecast iOS and Android OS release dates, addressing a critical business challenge. By leveraging time-series forecasting and deep learning techniques, the solution ensured timely application releases, minimized post-release issues and enhanced operational efficiency.

- Built an **LSTM-based time-series forecasting** model using **Python and TensorFlow** to predict iOS and Android OS release schedules, improving model accuracy from **76% to 91%**.
- Analyzed historical release patterns to uncover critical trends driving release timing, refining data by addressing outliers and seasonality, and achieving a **20% improvement in reliability** for predictive modeling.
- Developed and deployed a **Flask-based API**, containerized with **Docker**, and hosted on **AWS ECS (EC2 mode)** with **ALB**, ensuring high availability, auto-scaling, and real-time predictions. Utilized **AWS ECR** for image versioning, **S3 for model storage**, and **CloudWatch** for performance monitoring and proactive issue detection.
- Tested and validated medical devices, reducing **analysis time by 40%** while ensuring compliance with safety and reliability standards.

VARUN ENTERPRISES

Data Scientist

Bangalore, India

Sept 2020 – April 2021

Client: IRA (Instrumental Research Associate)

Addressed the challenge of detecting micro-defects in products that were not visible to the naked eye. Developed and deployed a deep learning-based detection solution on a cloud platform for real-time analysis. This resulted in improved accuracy, reduced errors, and increased revenue

- Developed a **custom Convolutional Neural Network (CNN)** achieving **93%** accuracy for detecting defects in M4 and M12 hexagonal nuts.
- Conducted detailed **EDA** to analyze data patterns and improve quality, and pre-processed **2,000+ images** by standardizing dimensions, enhancing image quality, and increasing data diversity, ensuring robust input for model training.
- Implemented a **Flask-based** application deployed on **AWS EC2**, ensuring real-time defect detection with a response time of **~2 seconds**, optimized to match manufacturing line requirements for immediate detection without delays.
- Reduced **false positives from 50% to 25%** by fine-tuning the classification threshold, resulting in a 12% increase in revenue.

HCL TECHNOLOGIES

Software Engineer

Bangalore, India

Dec 2018 – Dec 2019

Client: Citi Bank

Resolved critical issues in trading applications, and optimized SQL queries to resolve FO issues, ensuring seamless operations and improved client satisfaction.

- Resolved critical issues in trading applications, achieving a **95% resolution** rate within 24 hours, ensuring seamless operations and improved client satisfaction.
- Optimized **SQL queries**, reducing system downtime by 20% and improving system performance, enabling faster trade execution and data retrieval.
- Enhanced database performance by fine-tuning queries and indexing strategies, reducing query execution time by **30%** and improving reporting efficiency.
- Leveraged SQL and data analysis techniques to identify bottlenecks and streamline workflows, leading to improved operational efficiency and system stability.
- Gained hands-on experience in SQL, query optimization, and data-driven problem-solving, building a strong foundation for advanced analytics roles.

PROJECTS

PRS Neurosciences

Multimodal LLM-powered Medical Image Analysis([Github](#))

To improve the speed and accuracy of brain MRI analysis, I developed a diagnostic question-answering (QA) system in collaboration with a medical professional. By fine-tuning multimodal language models and implementing advanced preprocessing, I delivered an interactive QA tool that empowered doctors with faster real-time diagnostic insights.

- **Collaborated with medical professionals** to fine-tune advanced **multimodal large language models** (LlaVA_Med, LlaVA, and IDEFICS2), ensuring high accuracy and actionable insights for medical diagnostics. Achieved the highest performance with **LlaVA_Med**, optimizing its capabilities for enhanced clinical decision support.
- Designed and implemented an advanced preprocessing workflow to enhance image quality, standardize data, and extract critical features, while improving the consistency and relevance of text data, ensuring optimized inputs for superior model performance.
- Achieved a **loss of ~0.0034** and a BLEU score of **0.85** post-fine-tuning, validating the model's ability to generate accurate, natural language diagnostic responses while reducing MRI analysis time significantly.
- Delivered a **Gradio-based interactive tool** that reduced MRI **analysis time by 30%**, enabling medical professionals to upload MRI scans, receive precise, real-time diagnostic answers, and significantly improve their workflow efficiency in critical cases.

End-to-End MLOps Project with ETL Pipeline- Network Security System(Github)

Designed and implemented an end-to-end network security system to detect and mitigate threats in real time. Optimized data processing, model performance, and deployment workflows to enhance detection accuracy and efficiency. Automated the entire pipeline, ensuring seamless scalability and reliability for continuous threat monitoring.

- Developed **ETL** pipelines with **MongoDB Atlas** for efficient **big data processing**, reducing data inconsistencies by 20% and improving data integrity.
- Optimized **threat detection** models using **hyperparameter tuning and MLFlow**, increasing detection accuracy by 35% while streamlining model experimentation.
- Automated **deployment workflows** by integrating **Docker, GitHub Actions, and AWS EC2**, ensuring scalable, real-time threat detection with minimal downtime.
- **Implemented CI/CD pipelines and cloud-based model hosting**, reducing deployment time by 40% and enabling continuous monitoring for performance optimization.

SKILLS

- **Languages:** Python, SQL, R
- **Framework/Libraries:** TensorFlow, PyTorch, LangChain, Spark, Streamlit, Keras, Plotly, Pandas, NumPy, Scikit-Learn, NLTK
- **Machine Learning:** Logistic and Linear Regression, Random Forest, XGBoost, SVM, K-Means, PCA
- **MLOps:** CI/CD, Docker, MLflow, Kubeflow, Model Monitoring, Version Control (Git)
- **Cloud:** AWS (SageMaker, S3, EC2, ECR)
- **Databases:** MongoDB, MySQL
- **Visualization:** Tableau, Power BI, Excel
- **Deep Learning:** ANN, CNN, RNN, LSTM, Transformers, Computer Vision, Natural Language Processing
- **Mathematics:** Statistics, Probability, Calculus, Linear Algebra, A/B testing, Hypothesis Testing

EDUCATION

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| ● UNIVERSITY OF MASSACHUSETTS, DARTMOUTH
<i>MS in Data Science (GPA 3.97 / 4)</i> | Massachusetts, US
Sep 2022 - Aug 2024 |
| ● NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY
<i>B.E, Electronics & Communication (GPA 3.1 / 4)</i> | Bangalore, India
Aug 2014 - Jul 2018 |